

Hands-on 4 : Autoencoders and latent space representation



— Montreal, June 15–19 —

DLM I 2026

Pierre-Marc Jodoin / Nathan Painchaud / Thierry Judge

Fonds de recherche
Nature et
technologies

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Sherbrooke

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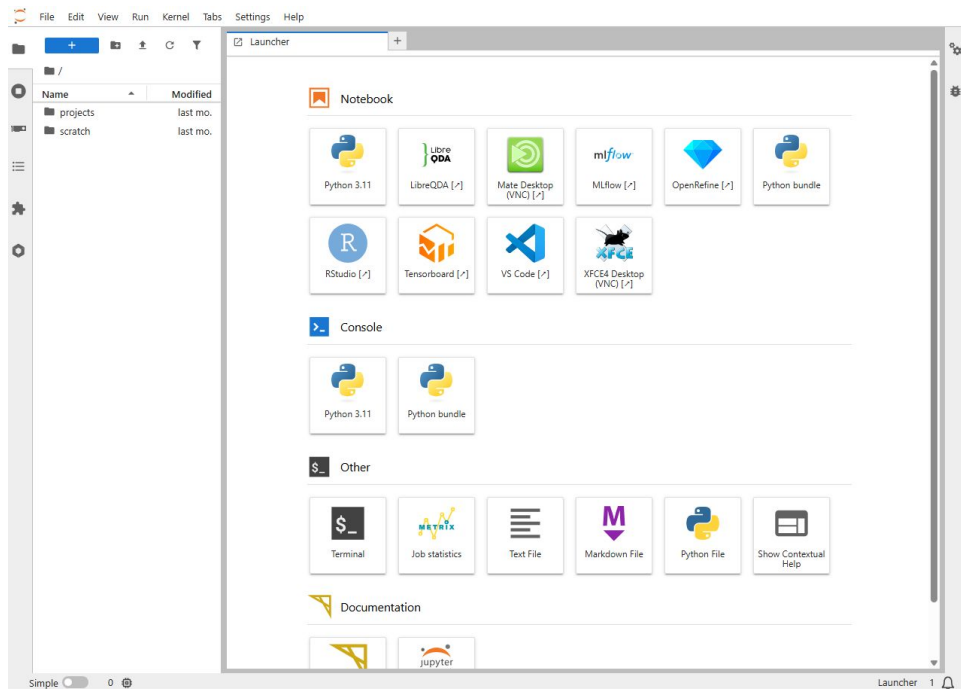
CREATIS

 UNIVERSITÉ DE
SHERBROOKE

Start a session

As for the previous hands on session, long onto the CalculQuebec environment and start a compute node

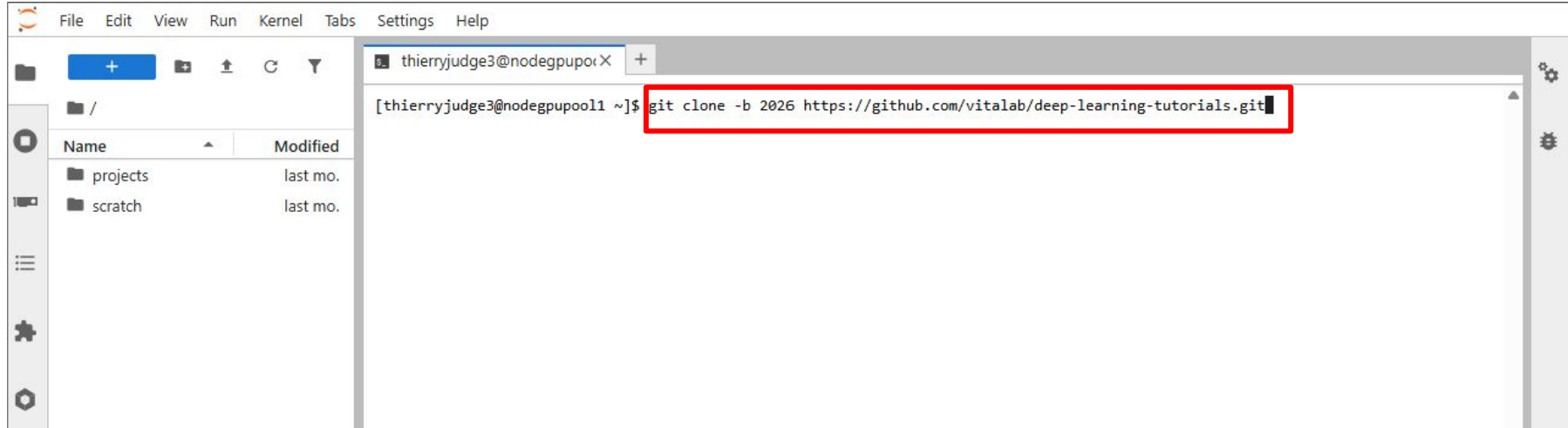
<https://jupyter.dmlti.calculquebec.cloud/>



Clone the repository

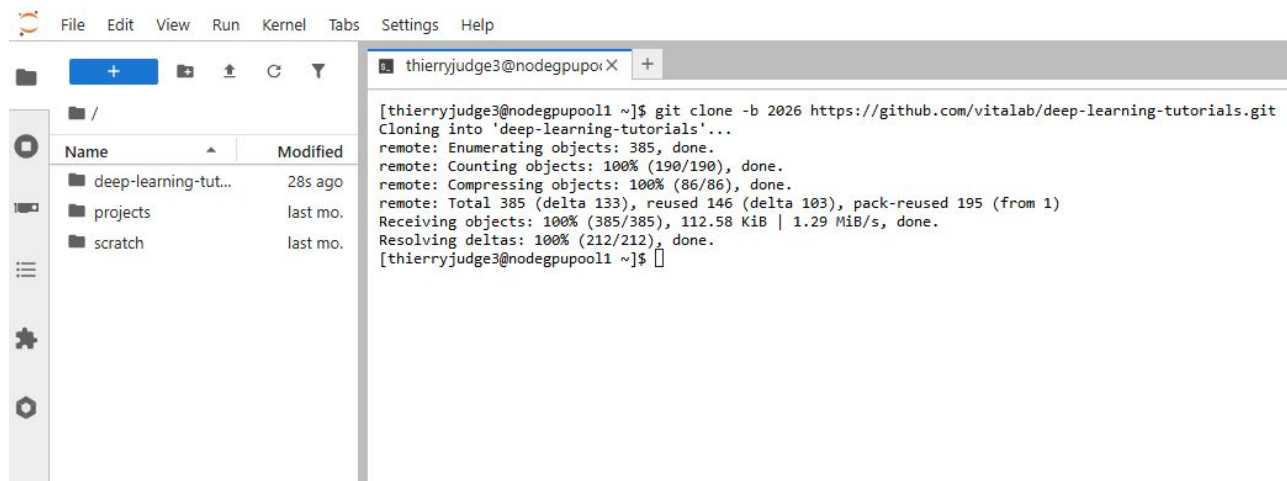
In the terminal, copy the following command and enter

`git clone -b 2026 https://github.com/vitalab/deep-learning-tutorials.git`



Clone the repository

You should see the following output in the terminal and a new folder should appear on the left. If you do not see the folder click on **refresh**



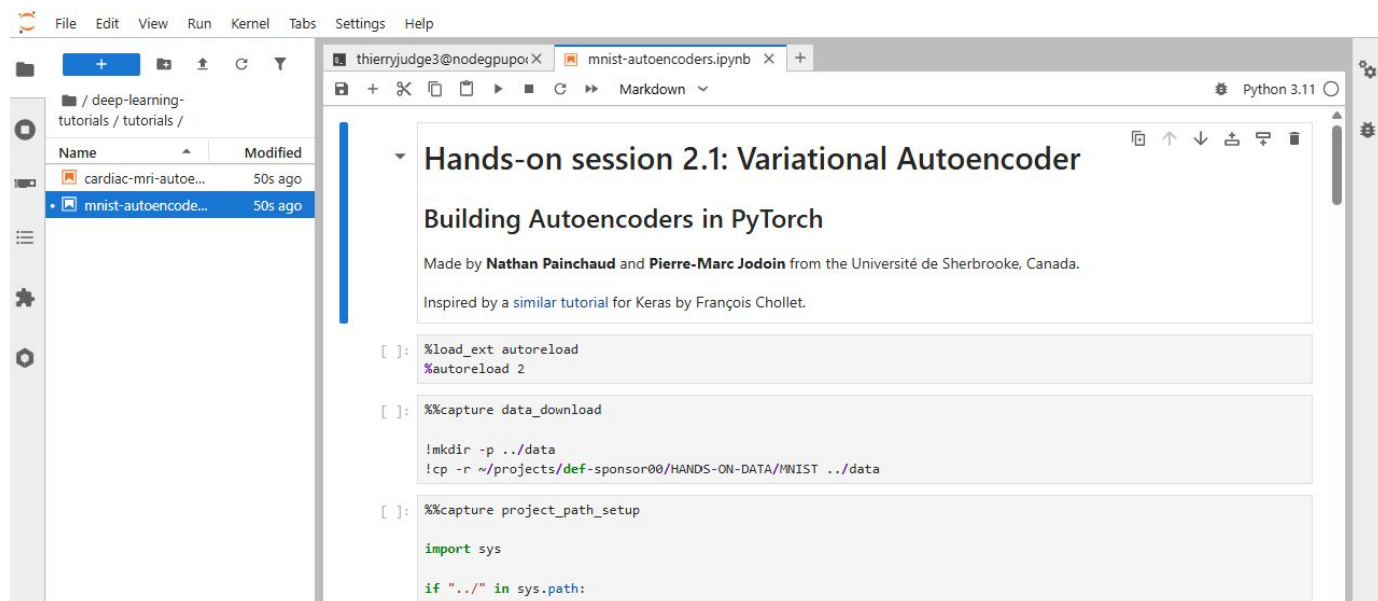
The screenshot shows a JupyterLab environment. On the left, the file browser displays the root directory with three folders: 'deep-learning-tut...' (created 28s ago), 'projects' (created last month), and 'scratch' (created last month). On the right, a terminal window titled 'thierryjudge3@nodegpupool1' shows the output of a git clone command. The output indicates that the repository was successfully cloned from GitHub.

```
[thierryjudge3@nodegpupool1 ~]$ git clone -b 2026 https://github.com/vitalab/deep-learning-tutorials.git
Cloning into 'deep-learning-tutorials'...
remote: Enumerating objects: 385, done.
remote: Counting objects: 100% (190/190), done.
remote: Compressing objects: 100% (86/86), done.
remote: Total 385 (delta 133), reused 146 (delta 103), pack-reused 195 (from 1)
Receiving objects: 100% (385/385), 112.58 KiB | 1.29 MiB/s, done.
Resolving deltas: 100% (212/212), done.
[thierryjudge3@nodegpupool1 ~]$
```

One the first notebook

Using the menu on the left, navigate to **deep-learning-tutorials/tutorials**.

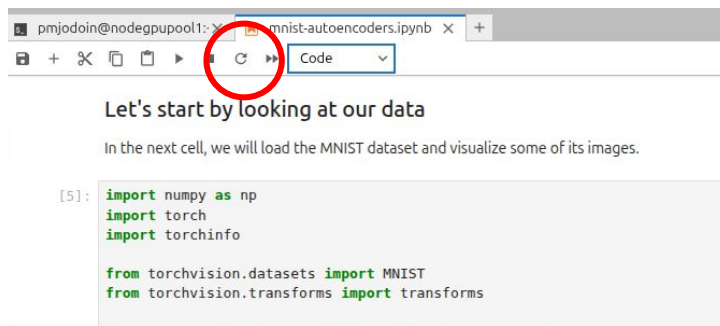
Click on `mnist-autoencoders.ipynb`. This will open the notebook.



If you see an error like this one:

```
-----  
ModuleNotFoundError                                Traceback (most recent call last)  
Cell In[5], line 3  
    1 import numpy as np  
    2 import torch  
----> 3 import torchinfo  
      4  
      5 from torchvision.datasets import MNIST  
      6 from torchvision.transforms import transforms  
  
ModuleNotFoundError: No module named 'torchinfo'
```

Restart the kernel by clicking on the circular arrow and go back to the first cell

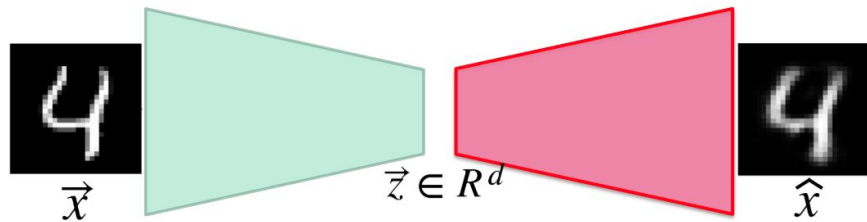


In this notebook you will learn...

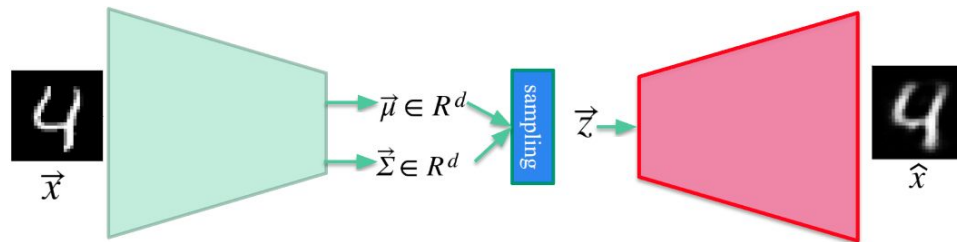
- What an Autoencoder is
- What a Variational Autoencoder is
- What a latent space is
- How you can use an Autoencoder to generate new MNIST data

In this notebook you will learn

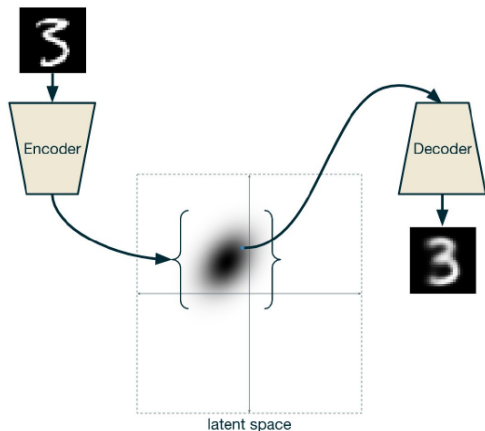
Autoencoder



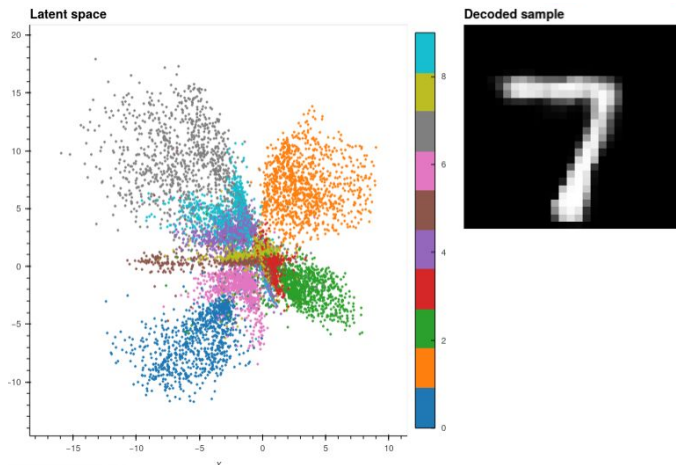
Variational Autoencoder



Latent space



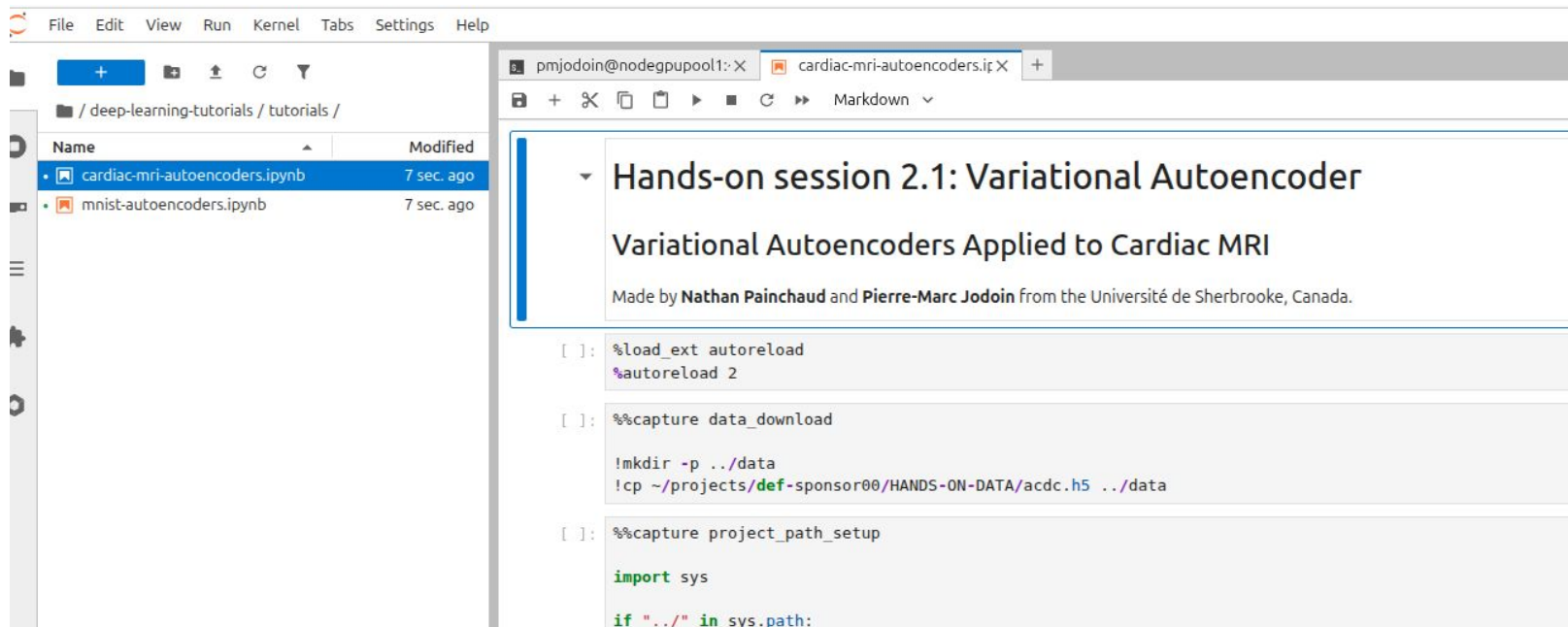
Generative model



The second notebook

Using the menu on the left, navigate to **deep-learning-tutorials/tutorials**.

Click on `cardiac-mri-autoencoders.ipynb`. This will open the notebook.



The screenshot displays the JupyterLab environment. On the left, the file browser shows the directory structure: `/ deep-learning-tutorials / tutorials /`. Two files are listed: `cardiac-mri-autoencoders.ipynb` (modified 7 sec. ago) and `mnist-autoencoders.ipynb` (modified 7 sec. ago). The `cardiac-mri-autoencoders.ipynb` file is selected. On the right, the notebook editor shows the title "Hands-on session 2.1: Variational Autoencoder" and subtitle "Variational Autoencoders Applied to Cardiac MRI". Below the title, it states "Made by Nathan Painchaud and Pierre-Marc Jodoin from the Université de Sherbrooke, Canada." The notebook content includes three code cells:

```
[ ]: %load_ext autoreload
      %autoreload 2

[ ]: %%capture data_download

      !mkdir -p ../data
      !cp ~/projects/def-sponsor00/HANDS-ON-DATA/acdc.h5 ../data

[ ]: %%capture project_path_setup

      import sys

      if "../" in sys.path:
```

On the second notebook

In this notebook, you will learn how to

- Apply an Autoencoder to medical imaging data
- Learn a cardiac shape latent space
- Generate synthetic cardiac shapes

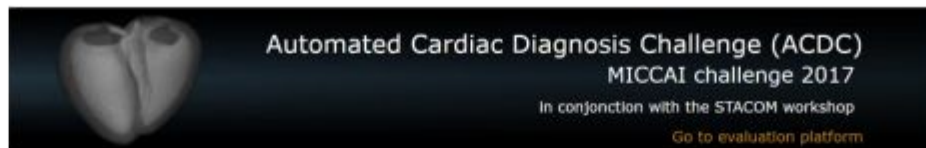
Automated Cardiac Diagnosis Challenge



150 exams (all from different patients) divided into 5 groups:

- dilated cardiomyopathy (DCM),
- hypertrophic cardiomyopathy (HCM),
- myocardial infarction (MINF),
- abnormal right ventricle (RV)
- patients without cardiac disease (NOR).

website: ***creatis.insa-lyon.fr/Challenge/acdc***



Overview
Contest
Participation
Databases
Evaluation
Code
Paper Submission
Contact

Overview

[General context](#) [Scientific interests](#) [Organizers](#)

The goal of this contest is two-fold:

- compare the performance of automatic methods on the segmentation of the left ventricular endocardium and epicardium as the right ventricular endocardium for both end diastolic and end systolic phase instances;
- compare the performance of automatic methods for the classification of the examinations in five classes (normal case, heart failure with infarction, dilated cardiomyopathy, hypertrophic cardiomyopathy, abnormal right ventricle).

This will be done using a common database of 3D cine-MRI images acquired from 150 patients (30 per pathology plus 30 healthy subjects) and the associated manual references based on the analysis of a clinical expert.

NEWS

4th of April 2017

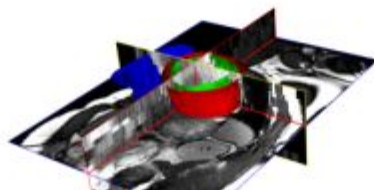
The full training dataset (100 patients) is available

13th of March 2017

Challenge accepted by the MICCAI Satellite Committee

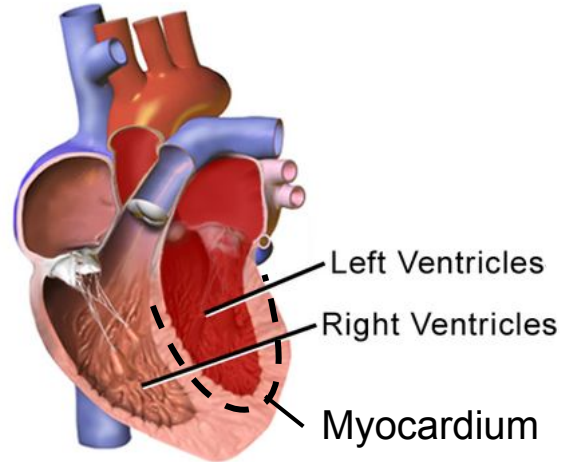
10th of January 2017

The online evaluation platform is operational



Cardiac anatomy

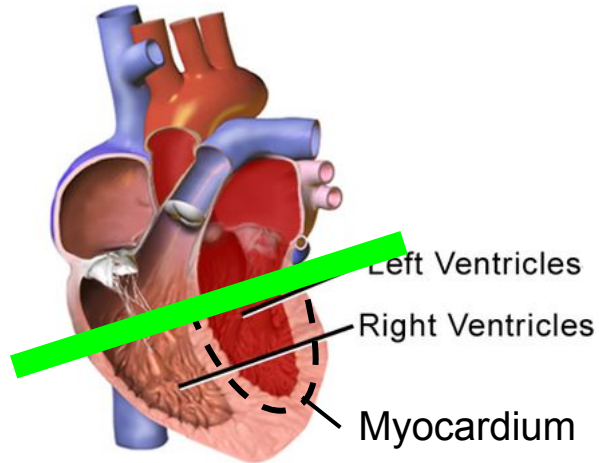
Normal Heart



Chambers relax and fill,
then contract and pump.

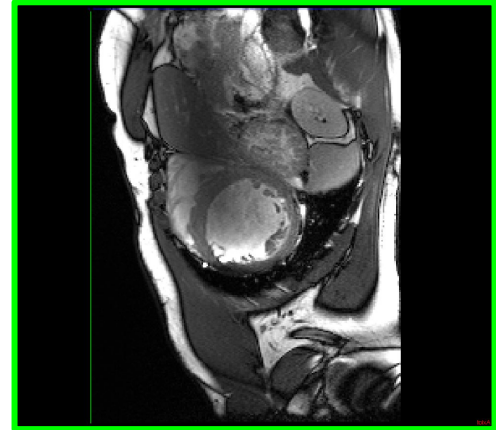
Cardiac anatomy

Normal Heart



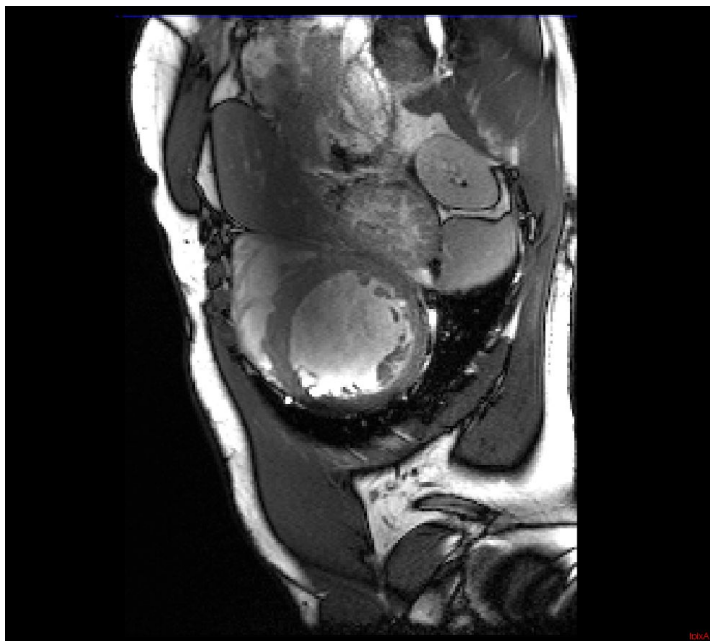
Chambers relax and fill,
then contract and pump.

Cross-section : **short axis view**

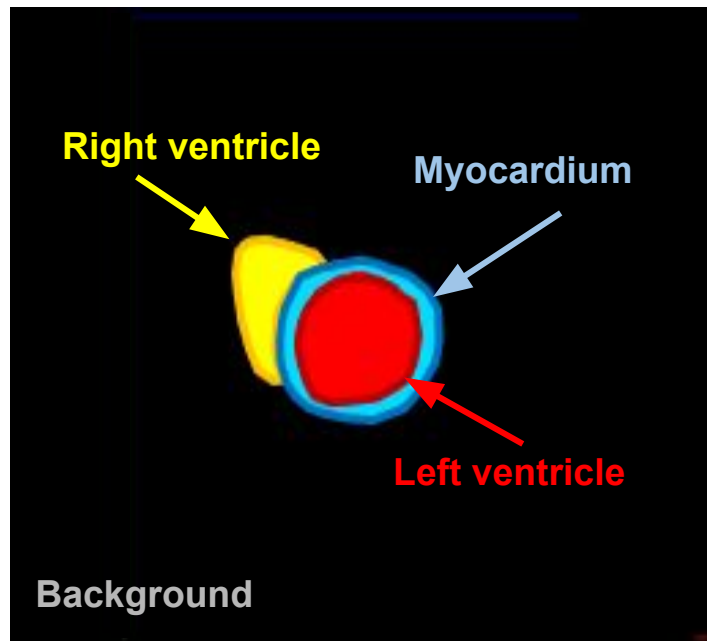




Short axis cardiac MRI

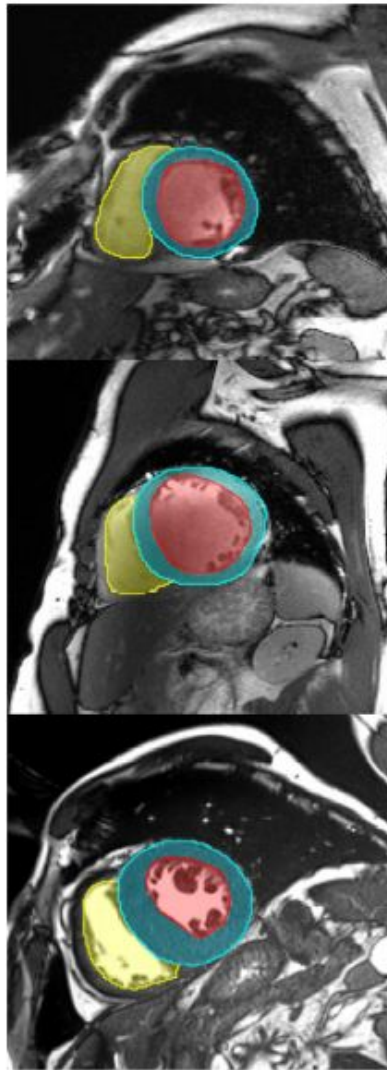


Short axis segmentation map





Other examples:



Convolutional Variational autoencoder

